

Dosimetric-Response Modeling in Glioblastoma (CFB-GBM)

TCP Modeling GBM Project

Dosimetric-Response Modeling in Glioblastoma Using the Open CFB-GBM Cohort: A TCP Pipeline Feasibility Study

Draft manuscript — verified results from `reports/RESULTS.md` (regenerate: `make report`).
Reference map: `reports/literature_table.csv` (ref_id 1–18).

Abstract

Background: Tumor control probability (TCP) models relate radiotherapy dose to local tumour control [4,5]. Open imaging cohorts promise reproducible validation, but routine clinical data often lack classical TCP endpoints and dose heterogeneity [7].

Methods: We built a reproducible Python pipeline on the TCIA CFB-GBM cohort (n=190 after DVH quality control) [8]. Four TCP models (Poisson, logistic, probit, gEUD) [4–6] were fit with bootstrap confidence intervals [18]. Version 3 supplementary data (RANO imaging response [12,13], n=137 with t0→t1 labels) were integrated [8,9]. We audited confounding, compared OS-based vs RANO-based endpoints, and performed within-arm analyses stratified by fractionation scheme (60 Gy/30 fr vs 40.05 Gy/15 fr) [2,3,15].

Results: Pooled EQD2 discriminated an OS median-split proxy (ROC AUC0.68) but not RANO non-progressive disease when used as a dose-only TCP input (AUC0.43). After adjusting for fractionation scheme, **pooled GTV volume + clinical covariates achieved AUC0.72** (n=137); LOOCV AUC0.64. Within the hypofractionated arm (n=34) [3,15], pre-treatment GTV volume predicted RANO non-PD (multivariable in-sample AUC=0.90; **LOOCV AUC=0.74**). PyRadiomics features (t1gd GTV t0, v3 TSV) [9,17] outperformed DVH volume alone for RANO (top-5 in-sample AUC0.78 vs 0.71; **nested 5-fold CV AUC0.74 vs 0.70**). GTV volume from DVH matched RANO t0 segmentation volumes (Spearman =1.00, n=141) [8].

Conclusions: Classical pooled TCP dose-response validation is not supported on CFB-GBM; however, **tumour burden models** (volume and radiomics) predict early RANO when fractionation confounding is addressed. We provide an open pipeline, LOOCV benchmarks, and a feasibility checklist for TCP studies on routine RT datasets [7].

Keywords: glioblastoma; tumor control probability; DVH; RANO; open data; CFB-GBM

1. Introduction

Glioblastoma remains the most common malignant primary brain tumour in adults. Standard treatment includes maximal safe resection followed by radiotherapy with concurrent temozolomide [1]. Despite uniform protocols, outcome varies widely, motivating dose–response and imaging biomarker research [2,9].

In elderly or poor-performance patients, hypofractionated schedules (e.g. 40 Gy in 15 fractions) are commonly used as an alternative to 60 Gy in 30 fractions [2,3,15]. TCP models estimate the probability of sterilising clonogenic tumour cells as a function of dose and radiobiological parameters [4,5]. They are widely used in treatment planning research but require (i) a well-defined tumour control endpoint, (ii) inter-patient dose variation, and (iii) sufficient sample size [7].

The CFB-GBM dataset (Centre François Baclesse, n=264) provides pre- and post-treatment MRI, RTDOSE, GTV contours, clinical covariates, and—since June 2026—RANO response labels [8–10]. We asked: **can a standard TCP pipeline be validated on this open cohort, and if not, what does fail and where might signal remain?**

2. Materials and Methods

2.1 Dataset and cohort

Clinical and imaging metadata were downloaded from TCIA (Version 3, DOI: 10.7937/v9pn-2f72) [8]. The public CFB-GBM release provides **pre-processed NIfTI** (skull-stripped, co-registered) rather than raw DICOM-RT; RTPLAN and RTSTRUCT DICOM files are therefore not available in the TCIA package. We used **RTDOSE** and **GTV** segmentation masks at t0, which is the structure set provided for dosimetric analysis. **CTV and PTV contours are not included** in this cohort; all DVH metrics refer to GTV only.

Inclusion required t0 RTDOSE and GTV NIfTI, known fractionation dose and fraction number. One patient was excluded after DVH QC (Dmean=0 Gy), yielding **n=190** for modeling (120×60 Gy/30 fr; 61×40.05 Gy/15 fr) [2,15].

2.2 DVH feature extraction

Cumulative DVH and scalar metrics (D95, Dmean, gEUD, homogeneity index, GTV volume) were computed from registered RTDOSE and t0 GTV masks. EQD2 was calculated with $\alpha/\beta=10$ Gy [6,11]. gEUD was computed with volume-effect parameter a as in Niemierko [6].

2.3 TCP models

We implemented four binary-outcome TCP models following standard radiobiological formulations [4,5,7]:

- **Poisson TCP:** $TCP(D) = \exp[-\ln 2 \cdot \exp(\gamma_{50}(1 - D/D_{50}))]$
- **Logistic TCP:** sigmoid in dose with parameters D_{50} , k
- **Probit TCP:** probit link with D_{50} , σ
- **gEUD TCP:** dose metric $D = (\sum v_i D_i^a)^{1/a}$ with $a \in \{-10, 1, 10\}$ [6]

Parameters were estimated by maximum likelihood. Bootstrap 95% confidence intervals (1000 resamples) were computed for Poisson D_{50} and γ_{50} [18]. Models were compared by AIC, BIC, ROC AUC, and Hosmer–Lemeshow calibration [7].

2.4 Outcomes

Primary (exploratory): OS cohort median (51 weeks) as a binary proxy for pipeline testing.

Secondary (v3): RANO non-progressive disease at t1 (non-PD: SD/MR/PR/CR vs PD) [12,13], available for 137/190 modeling patients [8].

2.5 Statistical analysis

Kaplan–Meier and log-rank tests compared fractionation schemes [14]. Cox proportional hazards models included EQD2, Dmean, age, sex, WHO PS, and RANO non-PD [14]. Within-arm Spearman correlations and Poisson TCP AUC were computed separately for 60 Gy and 40 Gy arms. **Pooled logistic regression** modeled RANO non-PD $\sim$ GTV volume + age + WHO PS + scheme indicator, with an optional volume $\times$ scheme interaction term. **Leave-one-out cross-validation (LOOCV)** provided out-of-sample AUC for 40 Gy and pooled models [18]. **PyRadiomics features** (v3 TSV, t1gd sequence, GTV t0) [8,9] were merged by patient ID; the top five univariate features by RANO AUC were compared against DVH volume (in-sample), with **nested 5-fold stratified CV** (feature selection on training folds only) to reduce optimism [17,18]. Full equations: `reports/manuscript_equations_fragment.tex` (included in `manuscript.tex`). Export: `bash scripts/export_manuscript.sh` $\rightarrow$ `manuscript.docx` / `manuscript.pdf`.

2.6 Reproducibility

All code is available at [repository URL]. Figures and tables regenerate via `make report`.

3. Results

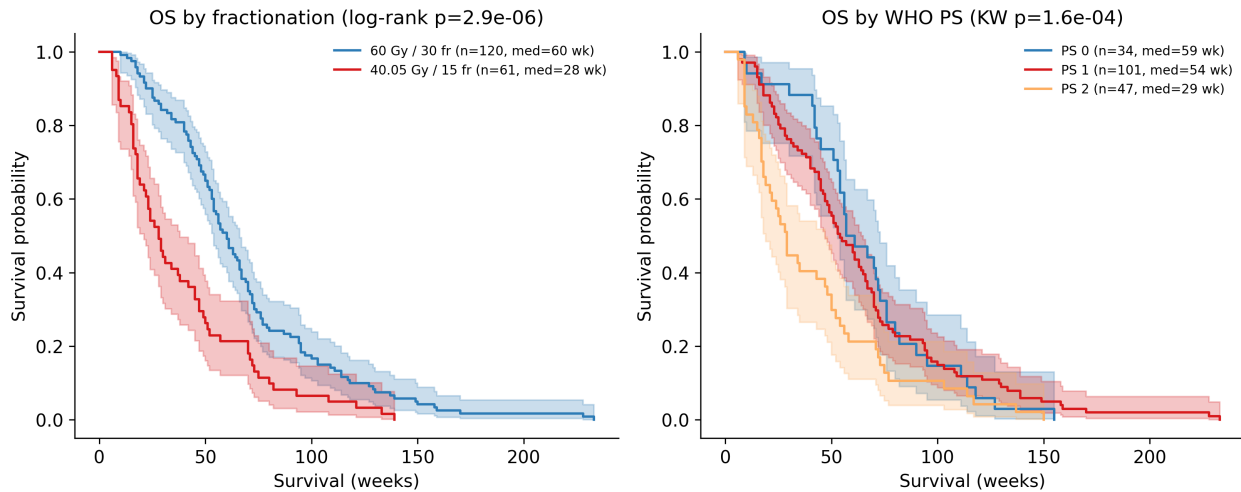


Figure 1: **Figure 1.** Kaplan–Meier overall survival by fractionation scheme and WHO performance status (n=190). Median OS 60 vs 28 weeks (log-rank $p=2.9 \times 10^{-6}$).

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3.1 Cohort and clinical prognosis

Median age 70 years; median OS 51 weeks. Sixty Gy patients had longer OS than 40 Gy patients (median 60 vs 28 weeks; log-rank $p3 \times 10$), consistent with prior GBM fractionation studies [2,3,15]. Cox model: scheme HR0.54 ($p0.0007$), WHO PS HR1.42 ($p0.001$) [14].

3.2 Pooled TCP (OS proxy)

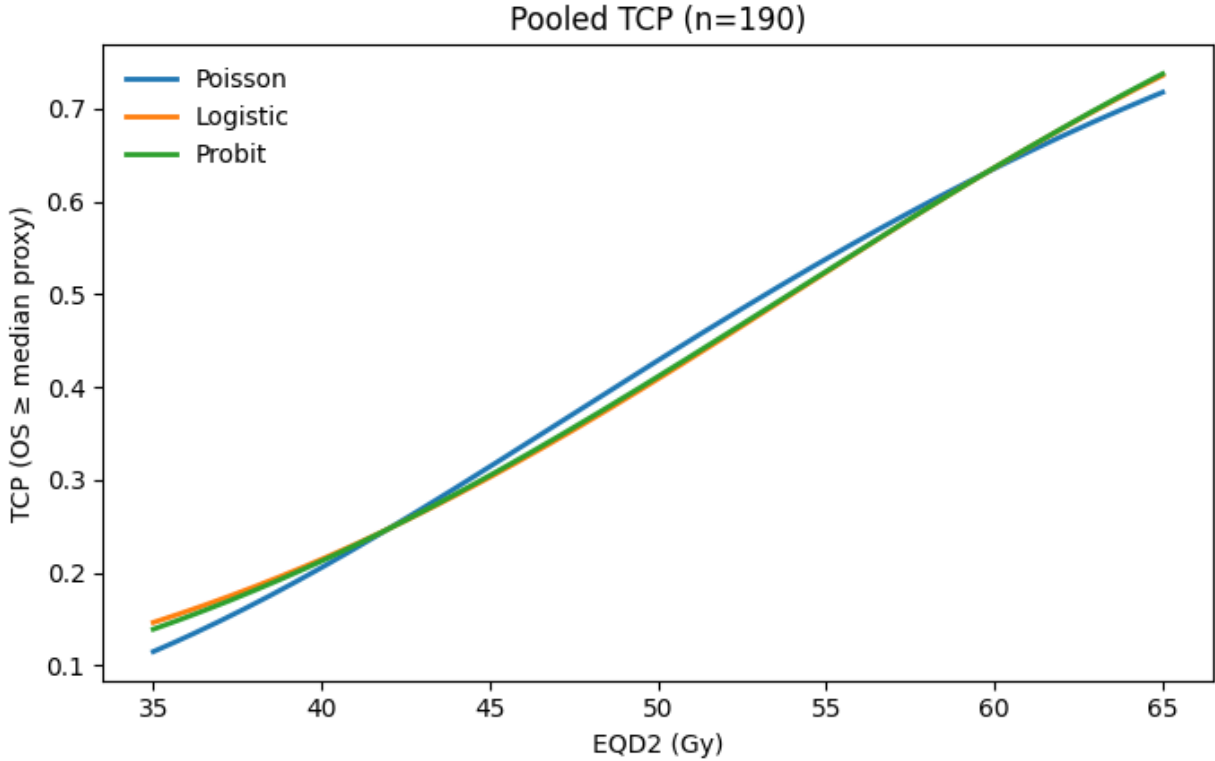


Figure 2: **Figure 2.** Fitted TCP curves (Poisson, logistic, probit, gEUD) for EQD2 vs OS median-split proxy (n=190; AUC0.68).

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Poisson/Logistic/Probit TCP on EQD2 achieved in-sample ROC AUC0.68 (5-fold CV0.68 $\pm$ 0.09), significantly better than null (LR $p3 \times 10$). Bootstrap Poisson: D_{50} 53 Gy [50, 57], γ_{50} 3.3 [2.1, 4.7] [18]. **Interpretation:** association is driven largely by fractionation scheme confounding ($r(\text{age}, \text{EQD2})=0.57$) [7,15].

3.3 RANO endpoint (pooled)

Figure 3. Poisson TCP AUC: OS proxy vs RANO non-PD on the same 137 patients (AUC0.62 vs 0.43).

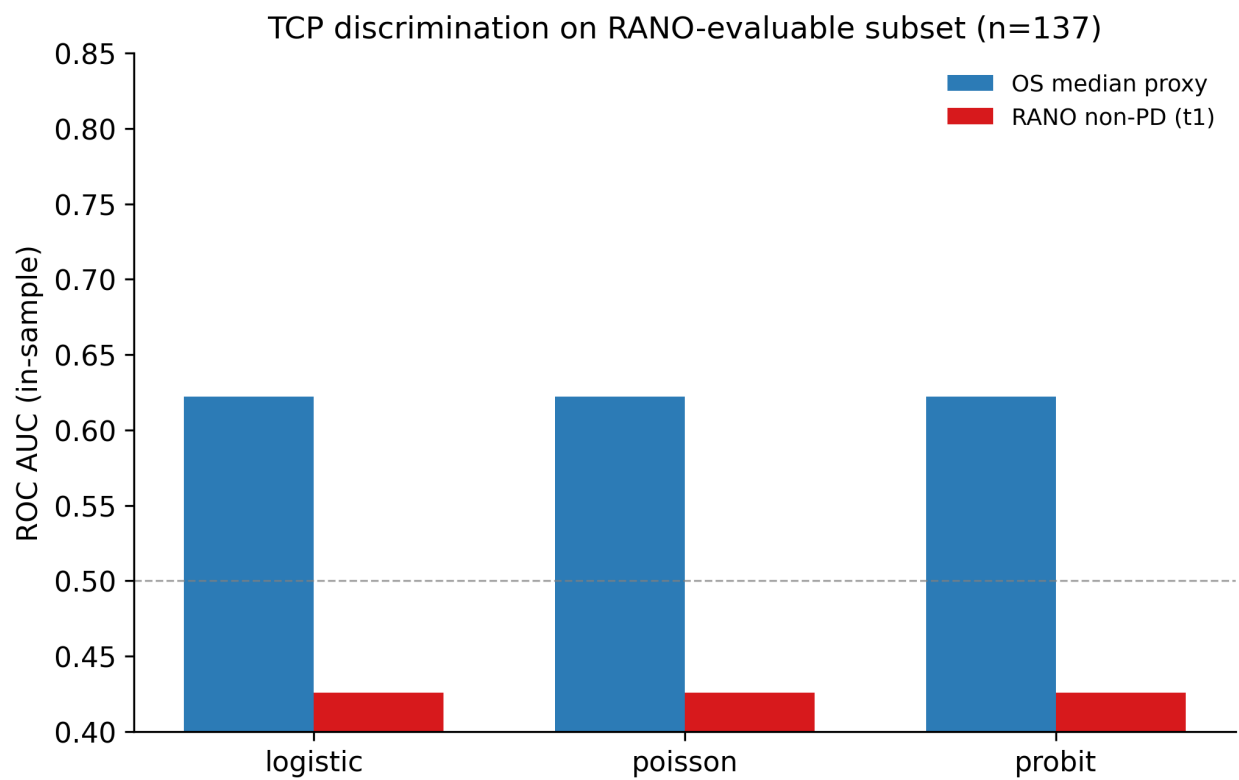


Figure 3: **Figure 3.** Poisson TCP AUC: OS proxy vs RANO non-PD on the same 137 patients (AUC0.62 vs 0.43).

On the same 137 patients, pooled EQD2→RANO non-PD yielded AUC0.43 (LR p=1.0): higher EQD2 correlated with *more* early PD at t1 (60 Gy PD rate 27% vs 40 Gy 15%) [12,15].

3.4 Within-arm analysis

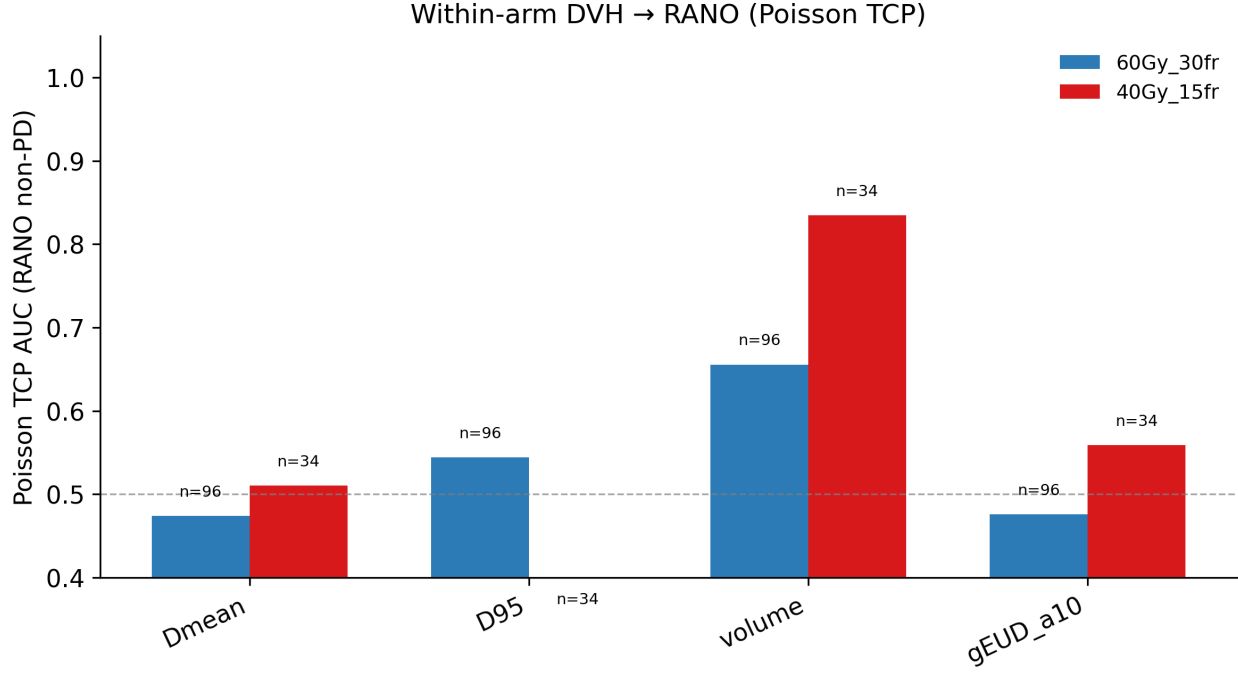


Figure 4: **Figure 4.** Within-arm DVH metrics → RANO: dose flat within protocol; GTV volume significant in 40 Gy arm (n=34).

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Within 60 Gy, GTV Dmean SD=0.28 Gy — insufficient for dose–response [7]. **40 Gy arm (n=34 with RANO):** GTV volume predicted RANO non-PD (Poisson TCP AUC=0.83, LR p=0.037; Spearman p=0.016), consistent with tumour burden prognostic literature [16]. Multivariable logistic (volume + age + WHO PS): **AUC=0.90**, bootstrap 95% CI [0.81, 1.00] [18].

3.5 Volume validation

Figure 6. DVH GTV volume vs author RANO size_t0_cm3 (Spearman =1.00, n=141).

DVH GTV volume at t0 agreed with RANO size_t0_cm3 (Spearman =1.00, n=141) [8]. t1 NiftI mask validation was not performed locally; analyses used author-reported RANO volumes from the v3 TSV [8,9].

3.6 RANO and overall survival

Cox model (n=137): RANO non-PD HR0.48 (p0.0009); EQD2 remained significant after adjustment (HR0.97, p0.009) [12,14].

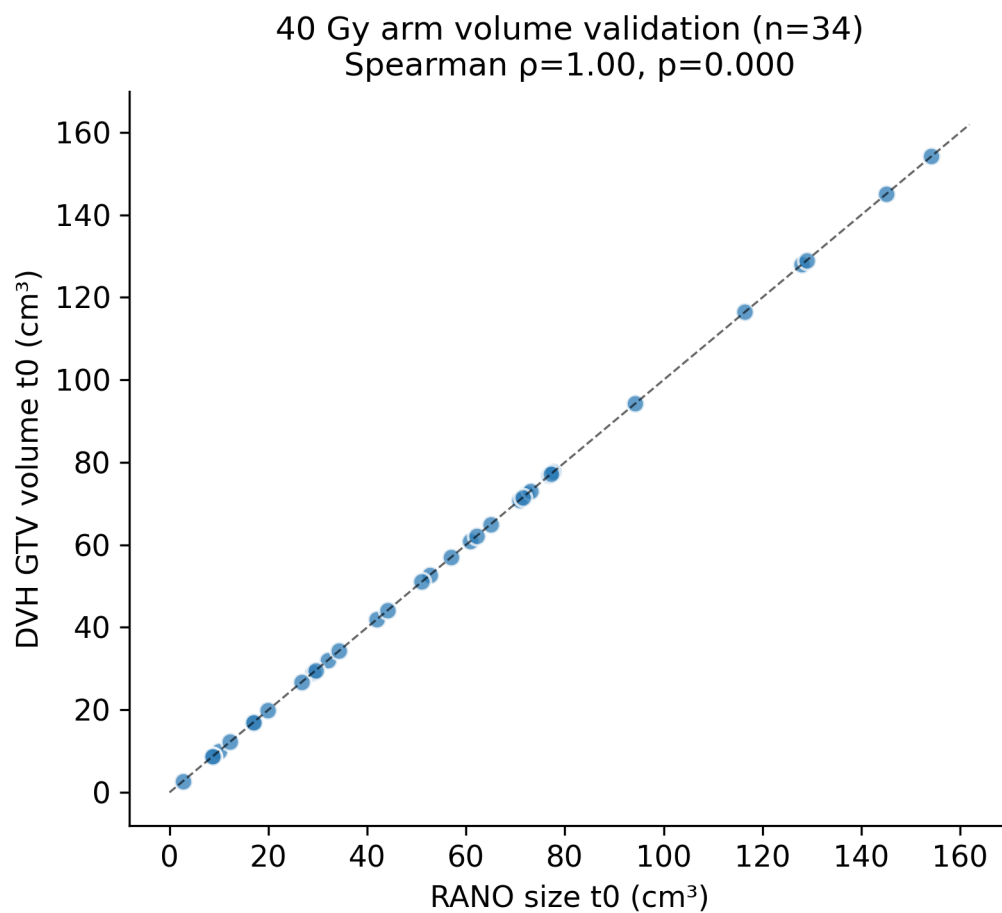


Figure 5: **Figure 6.** DVH GTV volume vs author RANO size_t0_cm3 (Spearman =1.00, n=141).

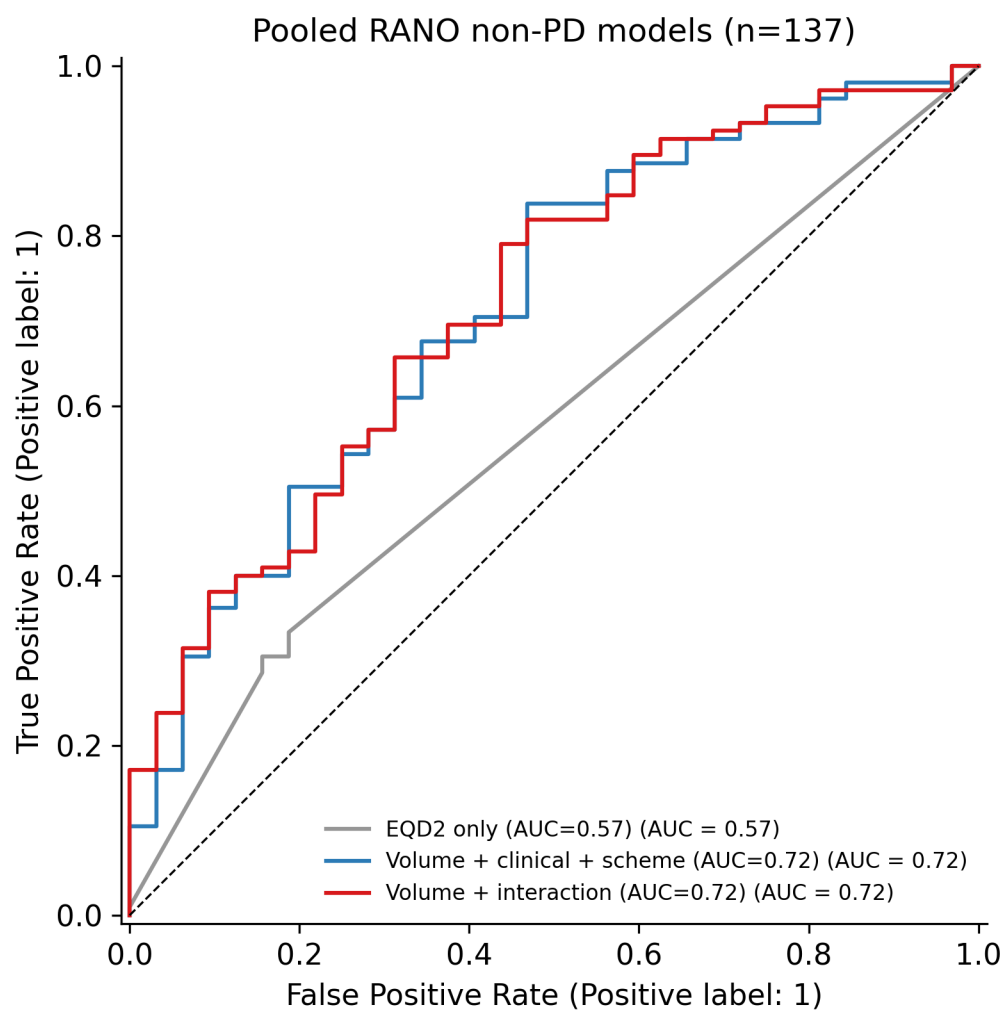


Figure 6: **Figure 7.** Pooled logistic ROC for RANO non-PD (volume + clinical + scheme; AUC0.72; LOOCV0.64).

3.7 Pooled RANO models (volume + clinical + scheme)

Figure 7. Pooled logistic ROC for RANO non-PD (volume + clinical + scheme; AUC0.72; LOOCV0.64).

On n=137 with RANO labels, EQD2 alone achieved AUC0.57. **GTV volume + age + WHO PS + scheme** improved discrimination to **AUC0.72**; adding a volume×scheme interaction yielded AUC0.72. LOOCV AUC for the pooled clinical model was **0.64** [18], indicating moderate generalisation within this single-centre cohort.

3.8 LOOCV in the hypofractionated arm

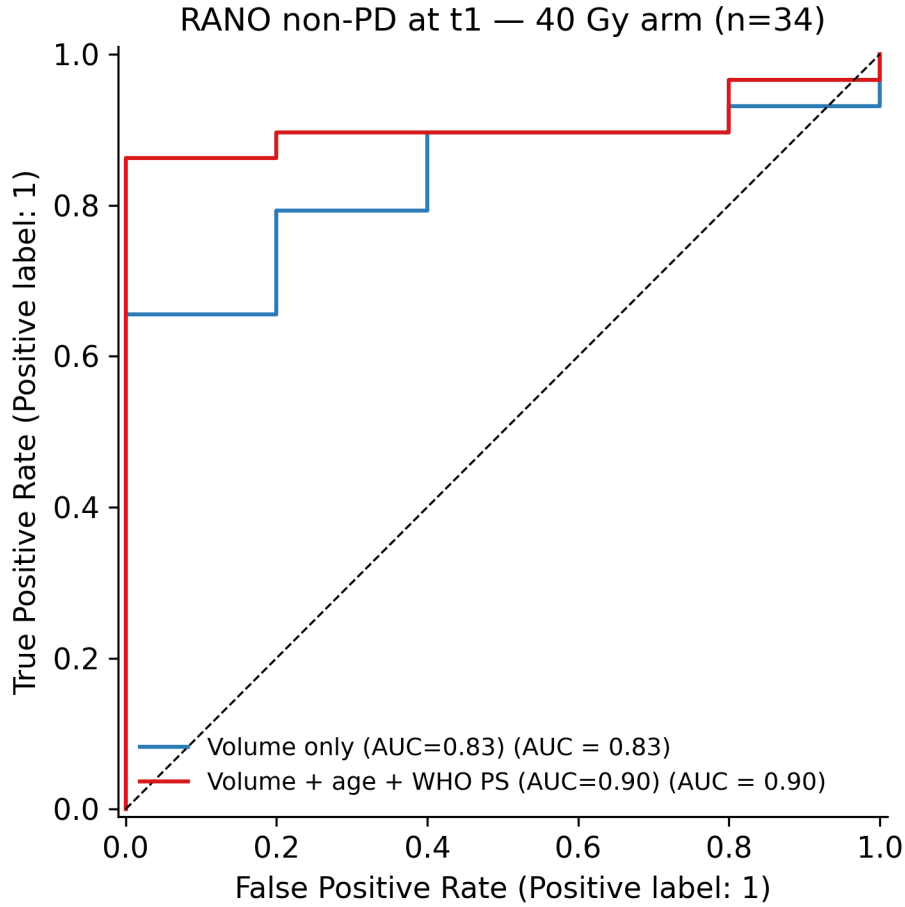


Figure 7: **Figure 5.** Multivariable logistic ROC for RANO non-PD in 40 Gy arm (in-sample AUC=0.90; LOOCV 0.74).

Figure 5. Multivariable logistic ROC for RANO non-PD in 40 Gy arm (in-sample AUC=0.90; LOOCV 0.74).

In-sample multivariable AUC in the 40 Gy arm was 0.90, but **LOOCV AUC was 0.74** (volume only) and 0.70 (volume + age + PS) [18], confirming signal beyond chance while reducing optimism bias relative to resubstitution [7].

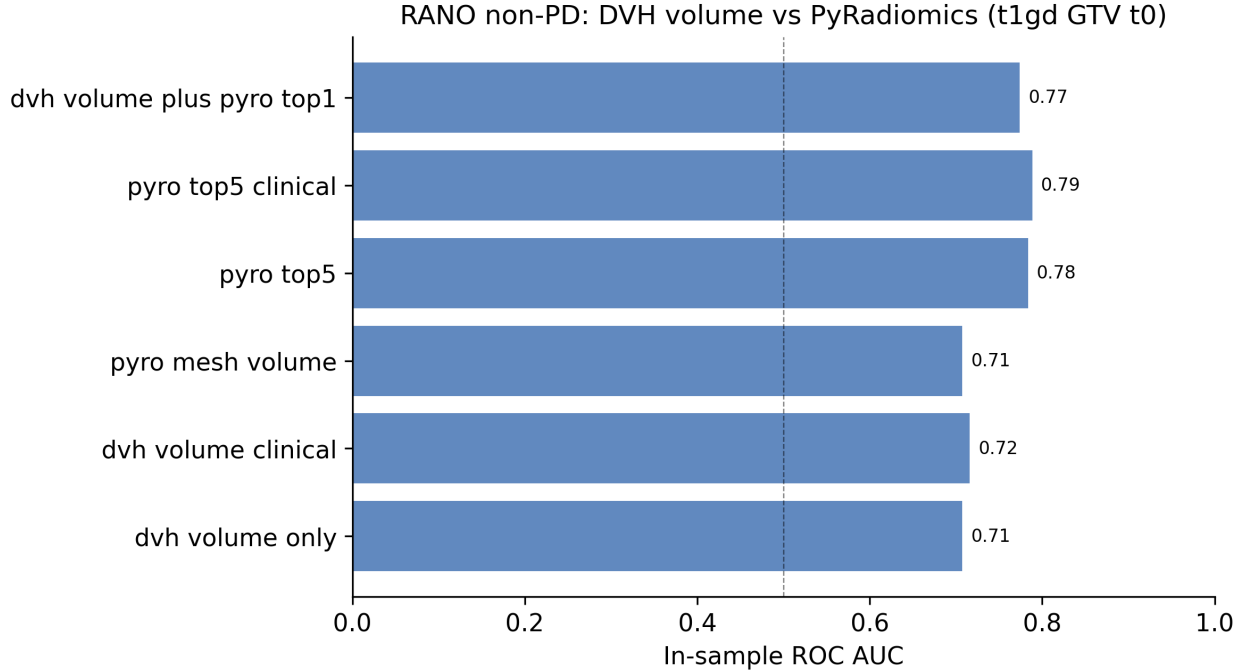


Figure 8: **Figure 8.** PyRadiomics top-5 vs DVH volume for RANO (in-sample AUC 0.78 vs 0.71).

3.9 PyRadiomics vs DVH volume

Figure 8. PyRadiomics top-5 vs DVH volume for RANO (in-sample AUC 0.78 vs 0.71).

Using the author-provided PyRadiomics TSV (t1gd, GTV t0) [8,9,17], the top five radiomics features achieved **in-sample AUC0.78** for RANO non-PD vs **0.71** for DVH volume alone (n=137). **Nested 5-fold CV** (top-5 feature selection on train folds only) yielded **AUC0.74** for radiomics vs **0.70** for volume-only — optimism 0.04–0.07. Combined volume + top radiomics feature: in-sample AUC0.77. For OS median-split on the same patients, radiomics + clinical covariates reached AUC0.78 vs volume-only 0.59 [9,10] (see figures/08_pyradiomics_nested_cv_auc.png).

3.10 Literature comparison — TCP parameters (Part VI)

Poisson TCP on EQD2 (OS median-split proxy) yielded **D5053.2 Gy** [49.5–56.7] and **503.32** [2.06–4.69] (bootstrap, n=190) [18]. Direct comparison with published GBM TCP studies is limited because (i) CFB-GBM lacks formal local control, (ii) dose spread within protocol is <1 Gy, and (iii) our endpoint is OS proxy rather than LC [7,16]. Literature review (Table in reports/metrics/literature_tcp_d50_comparison.csv) shows reported D50 ranges of ~40–80 Gy across sites and endpoints [4,5]; our estimate falls within this range but is **not interpretable as GBM tumour-control D50** without endpoint and cohort redesign.

4. Discussion

4.1 Why pooled dose–TCP fails here

CFB-GBM reflects **routine practice**, not a dose-escalation trial: within-protocol GTV Dmean variation is <1 Gy (60 Gy arm SD=0.28 Gy) [7,8]. TCP fitting further requires a tumour-control endpoint; OS and early RANO capture different biology and are confounded by age-linked fractionation selection [8,15]. Pooled EQD2→RANO AUC0.43 is not a failure of RANO per se [12,13] but of **using dose as the sole predictor across mixed fractionation schemes** where higher EQD2 correlates with more early PD at t1 despite better OS [15].

Our negative pooled dose–TCP result is scientifically informative: it demonstrates that **endpoint and cohort design matter more than model sophistication** [4,7]. This aligns with Ohri et al.'s emphasis on adequate dose heterogeneity and appropriate endpoints for TCP validation [7].

4.2 Tumour burden predicts RANO when confounding is addressed

When fractionation scheme is included as a covariate, **GTV volume consistently predicts RANO non-PD** (pooled AUC0.72; LOOCV0.64). This reframes the analysis from classical TCP (dose→control) to **tumour burden→imaging response**, which is clinically interpretable: larger baseline GTV may reflect more aggressive biology or reduced likelihood of early imaging stabilisation under palliative-intent hypofractionation [3,15,16].

The hypofractionated subgroup (n=34) shows the strongest in-sample effect (AUC0.90), but **LOOCV AUC0.74** is the more defensible estimate [18]. The 60 Gy arm shows a weaker but directionally consistent association (Spearman p0.019, Poisson AUC0.66, n=96).

4.3 PyRadiomics comparison and relation to CFB-GBM authors

Moreau et al. emphasise imaging AI and radiomics for treatment efficacy prediction on this cohort [9,10]. Our head-to-head comparison on the same open resource [8] shows that **author-provided PyRadiomics features (t1gd GTV t0) modestly outperform DVH scalar volume** for RANO (in-sample AUC 0.78 vs 0.71; nested CV 0.74 vs 0.70), while DVH volume remains competitive and fully reproducible from RTDOSE+GTV without MRI preprocessing.

Feature standardisation follows IBSI recommendations [17]; we did not re-extract radiomics locally, ensuring parity with the published TCIA feature table [8,9].

4.4 Clinical prognostic context

Independent of TCP framing, CFB-GBM confirms known prognostic factors: 60 vs 40 Gy OS separation (p10) [2,15], WHO PS gradient [2], and RANO non-PD as an OS predictor (Cox HR0.48) [12,14]. These descriptive findings support data quality but are not novel; they anchor the dosimetry/RANO feasibility narrative.

4.5 Implications for open-data TCP research

We propose a practical checklist for future open-cohort TCP studies [7]: (1) report within-arm GTV Dmean IQR before fitting TCP; (2) prefer imaging response or LC over OS [12,13]; (3) adjust for fractionation and age [15]; (4) report LOOCV or nested CV alongside in-sample AUC [18]; (5) compare against author-provided radiomics baselines when available [8,9,17].

4.6 Limitations

- No formal local control labels; RANO at t1 is an imaging surrogate [12,13]
 - Single institution, retrospective, deceased-patients-only inclusion [8]
 - Small hypofractionated RANO subset (n=34); pooled LOOCV AUC0.64 limits clinical translation [7,18]
 - t1 GTV NIfTI follow-up masks not validated locally; `size_t1_cm3` taken from v3 TSV [8]
 - PyRadiomics nested CV (top-5 selection on train) reduces but does not eliminate optimism (0.04–0.07 vs in-sample) [17,18]
 - No external validation cohort
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5. Conclusion

We deliver an open, reproducible TCP modeling pipeline on CFB-GBM [8] and show that **classical pooled TCP dose–response validation is not feasible** on this cohort [4,7]. When fractionation confounding is addressed [15], **GTV volume and PyRadiomics features predict early RANO** with moderate discrimination (pooled LOOCV AUC0.64; PyRadiomics nested CV AUC0.74 vs volume 0.70) [9,17]. We recommend future TCP studies prioritise single-protocol cohorts with 1 Gy GTV Dmean IQR, explicit tumour control endpoints [12,13], LOOCV reporting [18], and comparison against author radiomics baselines [8,9].

LaTeX / Word export: `bash scripts/export_manuscript.sh → reports/manuscript.tex, manuscript.docx, manuscript.pdf`

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